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RESEARCH

Using Archaeological Road Data to Evaluate Limits of Topography on Road Location: Roman Roads in the Near East as a Test Case

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Abstract

The article presents a new methodological approach to understanding (post-dicting) the location of ancient roads based on an inductive topographic analysis of existing road remains. The research is focused on the analysis of a spatially highly precise dataset of Roman roads in the Near East, using selected topographic variables (maximum slope, topographic position index, vector ruggedness measure). The identified limits of the topographic variables are used to create an isotropic (direction-independent) model of 'natural corridors of movement' using an adapted 'from everywhere to everywhere' (FETE) method. The modelled movement corridors are then overlaid by the known Roman road network in the Near East, and their overlap is calculated. The overlap between the modelled corridors and Roman roads is around ~ 25%, suggesting that the selected topographic variables and slope categories have limited explanatory power in understanding the layout of the Roman road network as a whole. The performance of the isotropic model is compared to four selected slope-based anisotropic functions (Tobler, Naismith, Herzog, Llobera-Sluckin). It is shown that the isotropic model incorporating a limited number of topographic variables better explains the location of Roman roads than the selected functions. It is argued that these results suggest that additional anthropic, cultural variables have a stronger influence on the layout of the Roman road network, and an alternative approach on how to incorporate topographic and anthropic variables is proposed.

Keywords Roman roads · Least cost path · Topographic variables · Movement corridors · Cost functions · Near East

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Introduction

The study of Roman roads recently moved beyond traditional historical-archaeological methods of enquiry (focusing on, e.g., ancient itineraries, milestones, and physical remains) into the realm of computational methods. Contemporary studies employ a wide array of spatial analytical methods in Geographic Information System (GIS) and most notably least-cost path (LCP) analyses (Herzog, 2020; Verhagen et al., 2019) for both predicting (Argüelles-Alvarez & Trapero-Fernández, 2024; Parcero-Oubiña et al., 2019) and post-dicting, i.e., explaining the locations of Roman roads (Arnoldus-Huyzendveld et al., 2016; Fonte et al., 2017; Güimil-Fariña & Parcero-Oubiña, 2015). Lewis (Lewis, 2021, 2024b) employs probabilistic LCP modelling to explain the location of Roman roads and to assess uncertainty of the modelled routes. These approaches are often concerned with modelling general movement patterns in the landscape and how these might relate to the evidence of Roman roads, forts and settlements (Lewis, 2024a; Parcero-Oubiña et al., 2023; Pažout, 2017). All these approaches are based either on 'traditional' LCP modelling using an array of variables to 'fit' course of a known road (Herzog, 2013b), or on 'direction-less' approach that aims to identify 'natural movement corridors' in the landscape (Déderix, 2017; Fábrega-Álvarez, 2006; White & Barber, 2012). LCP algorithms predominantly used nowadays are slope-based assuming that movement through terrain is either time-saving or energy-saving process and methods were developed to assess which might have played more role in decision making when establishing a path (Field et al., 2023). Preoccupation with slope means that there is often limited use of additional environmental, topographic, or anthropic variables (typically land cover or hydrology). And if so, selection of those variables is very often deductive, top-down process that incorporates existing archaeological data only in its last (verification) stage. Only Van Lanen et al. (2015) in the case of the Roman Netherlands incorporates wide range of environmental and anthropic variables to analyse potential for presence of roads in the landscape. What seems to be missing in both predicting and post-dicting approaches adopted in the aforementioned studies is the lack of bottom-up inductive process of analysis of available data for past movement that would be essentially road-oriented. Arguably, this was likely caused by lack of appropriate dataset of roads, although Roman roads are one the best documented remains of pre-modern transport infrastructure reflecting past human movement.

In this paper, I attempt first to analyse a spatially highly detailed dataset of Roman roads in the Levant (Fig. 1) to identify topographic constraints of the locations of the Roman roads and to use this data to build a general mobility model for the case study region in the southern Levant based on the notion of 'natural corridors of movement' (i.e., employing bottom-up inductive approach starting from available road dataset as a proxy for human movement). In turn, this mobility model is compared against the evidence for the Roman roads. By doing so it will be possible to evaluate the influence of the whole set of topographic variables on the location of Roman roads. The proposed model is isotropic, i.e., direction independent. Finally, the performance of this isotropic

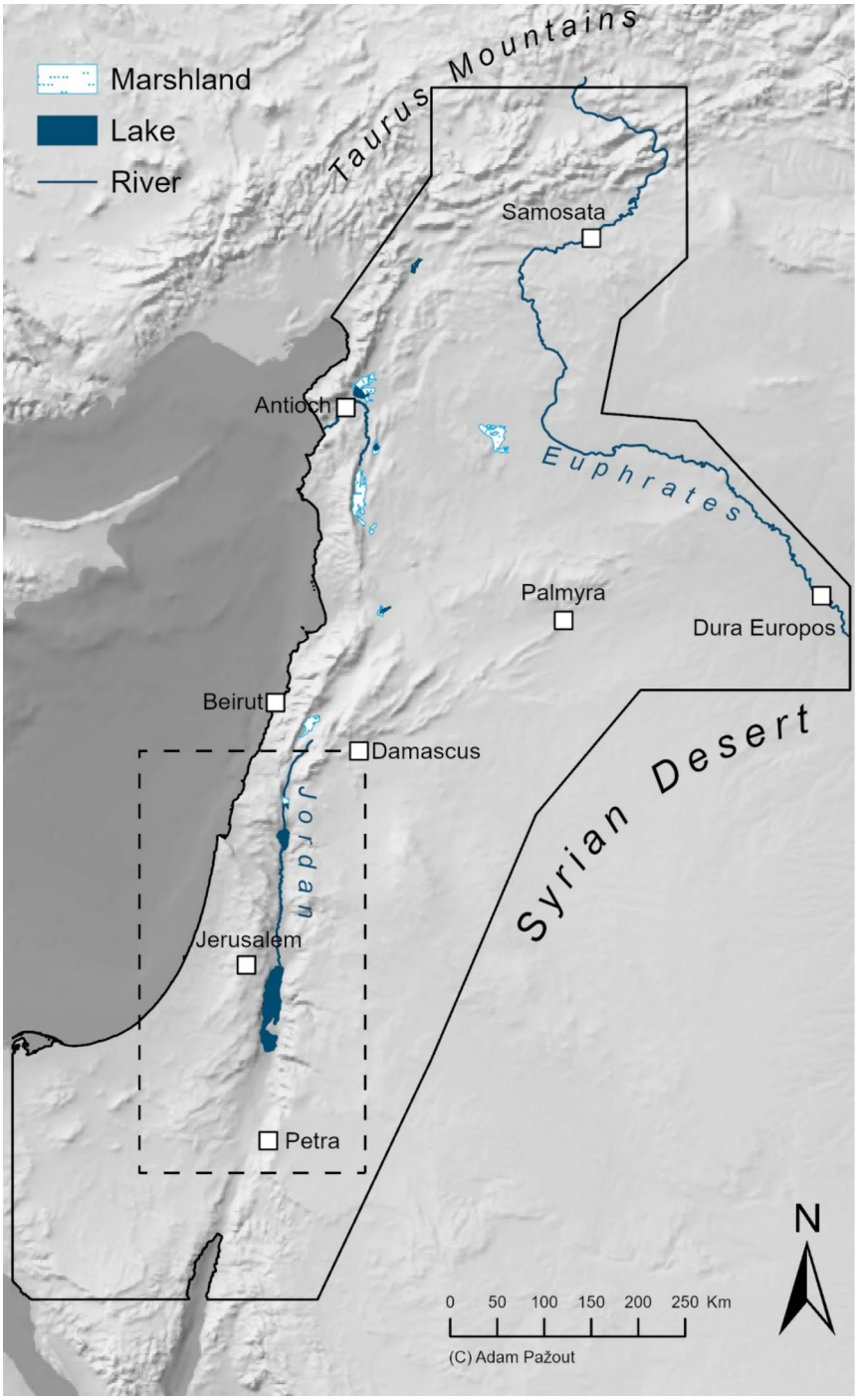

Fig. 1 Study region is outlined with a solid line; the dashed line indicates the case study in the southern Levant. Major Roman sites are shown

model in explaining the location of Roman roads is evaluated against selected time-saving and energy-saving slope-based algorithms. The principal question underlying this research is twofold: (1) is the isotropic model incorporating both slope categories and topographic variables better at explaining locations of Roman roads than slope-based models, and (2) how many locations of Roman roads can be explained solely by slope categories and topographic variables? By starting the enquiry with Roman roads as a proxy for the past movement patterns, it is hoped to avoid some pitfalls of the 'traditional' LCP algorithms described above and to gain empirically informed evaluation of the influence of topography on actual layout of the Roman road network.

Data and Methods

Data

The Roman road dataset for the Levant is derived from the *Itiner-e: The digital atlas of ancient roads* (Brughmans et al., 2024). It is based on the up-to-date published evidence for Roman roads across the whole of the Roman Empire. High spatial accuracy was achieved through a combination of modern and historical topographic maps, high-resolution satellite imagery, and modern orthophotos (De Soto et al., 2025). In total, 3877.8 km of roads in the dataset located in the Roman Levant (Fig. 2) belong to the 'certain' category, meaning that they were digitised with the highest spatial accuracy and resolution possible. Several smaller sections of roads were removed from the analysis since the modern terrain was modified by quarrying and building activity or flooded by reservoirs since the time of the identification of the roads. With 118.4 km such removed, the analysis is performed for a total of 3,759.4 km of roads found throughout the research area in various environments, although the southern Levant is more represented due to certain research bias (Pažout, 2025). For the same reason, the southern Levant was chosen as the case study since the Roman road network in the region is more representative to allow comparison with modelled 'natural corridors of movement'. To obtain a wide range of values for the topographic variables (see further), all roads were split into segments of maximum length of 1000 m, resulting in 3891 analysed segments (some shorter than 1000 m).

The Digital Elevation Model (DEM) used for the analysis is FABDEM (Forest and Building removed DEM, Neal & Hawker, 2023), which is based on the Copernicus GLO-30 DEM (European Space Agency and Airbus 2022). It provides the best resolution covering the whole research area at 1-arc second ca. 34 m at this latitude. Moreover, by removing the forest and building cover, it achieves higher vertical accuracy compared to similar global DEMs (Hawker et al., 2022) and better represents the underlying topography.

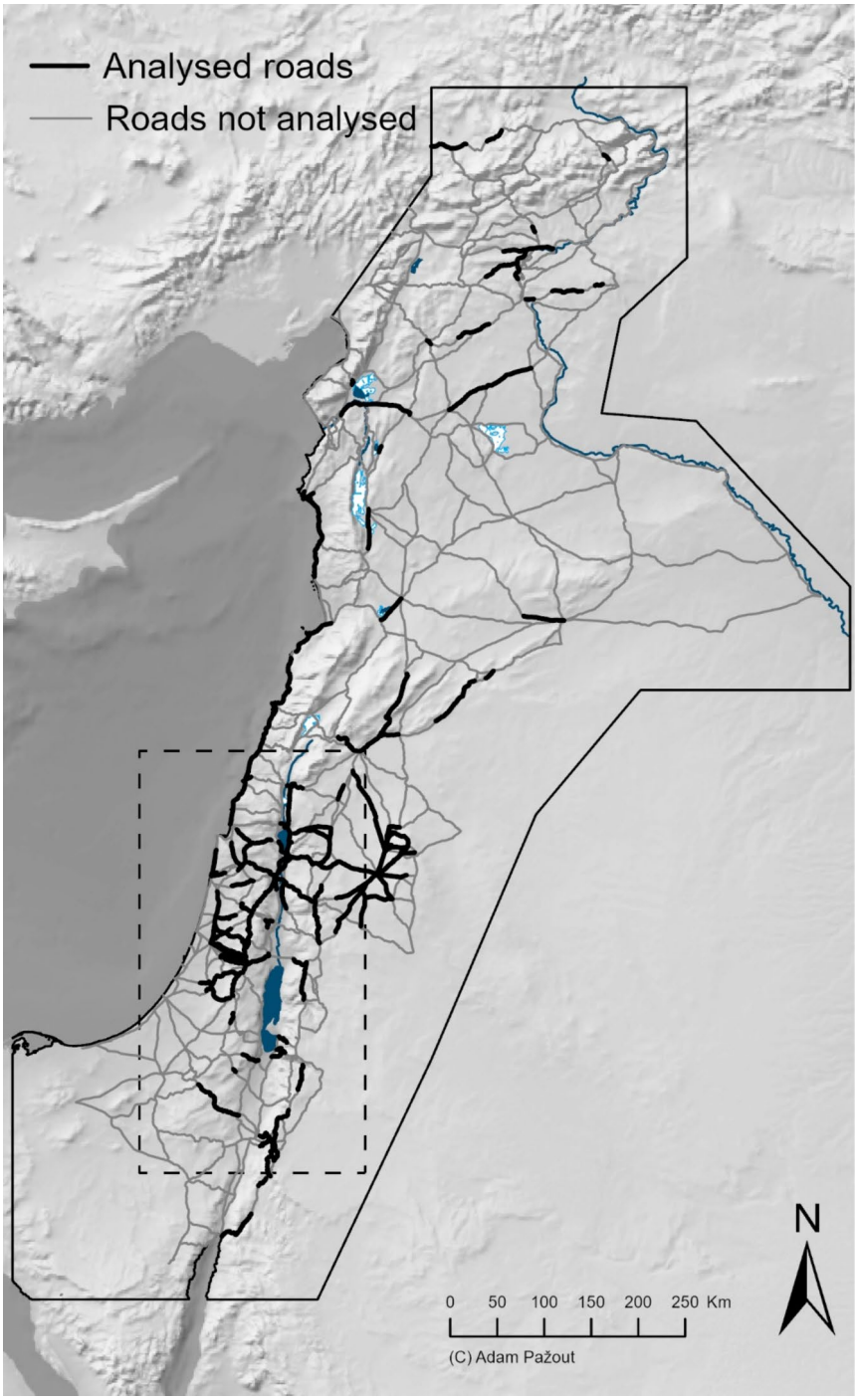

Fig. 2 Roman road network in the study region, showing roads used for the analysis of topographic variables ('certain' category)

Topographic Variables

For each of the 3891 road segments, these topographic variables are calculated: maximum slope, topographic position index (TPI), and vector ruggedness measure local (VRML). TPI is defined as the difference between the elevation of a focal cell and the mean elevation of its neighbourhood (Wilson & Gallant, 2000: 74). It gives an idea about the relative position of a cell within its neighbourhood, where negative values indicate valleys or lower slopes, positive values indicate upper slopes or ridges, and values around zero indicate flat terrain. VRML is a terrain metric derived from the terrain ruggedness index (mean of the absolute differences in elevations between a focal cell and its neighbourhood, Riley et al., 1999) that is designed to reduce the influence of slope and terrain curvature on the calculation (Dilts et al., 2023). TPI was calculated for a 33-cell neighbourhood (ca. 1100 m) so that it reflects the influences of local topography in the meso-scale (hillslope profiles, stream channels) rather than that of large landforms. VRML was calculated for a 3-cell neighbourhood to reflect terrain conditions on the micro-scale. Maximum slope was calculated using the *Add Surface Information* tool in ArcGIS Pro v3 (ESRI, 2024) and converted to degrees. TPI was calculated using GRASS GIS tools *r.neighbors* and *r.mapcalc;* VRML was calculated using *Topography Toolbox Pro* for ArcGIS Pro v3 (Dilts, 2023). Minimum, maximum, and mean values of TPI and VRML were summarised for each road segment using the *Add Surface Information* tool in ArcGIS Pro v3.

Assigning Conductivity Values

In order to calculate cost surfaces for the source points in the natural corridors of movement modelling (see the section), conductivity values need to be assigned to the conductance surface first. The conductivity is an inverse of friction, so that high cell values in the raster represent cells that are easier to cross. The conductance surface is used as it is employed in the *least-cost-path* package in R that was used in the calculations (Lewis, 2020). The values obtained from topographic analysis were reclassified into the conductivity surface based on the cutoff values identified in the previous step (see the “).

The values in the conductance surface were derived through a series of tests with different values for topography and critical slope. TPI and VRML cutoff values were combined into one raster, which was then reclassified into a conductance surface where the combined topographic variables are assigned arbitrary values (1, 2, 5, 10, 20, 50) and the surrounding surface is given value 100 (i.e., maximum conductivity). The same was made for critical slope, where cells above critical slope were assigned arbitrary values (5, 10, 20, 30, 50, 90) and surrounding surface value 100. LCPs were then modelled using these conductivity surfaces in a test case in the Jerusalem region on the roads between Jerusalem-Lydda, Jerusalem-Emmaus, and Emmaus-Lydda (Fischer et al., 1996). These three roads represent varied topography from flat coastal plain interspersed with shallow wide wadis, low rolling hills, to

steep ridges and deeply incised valleys. The topography and critical slope conductance rasters were first evaluated individually and then in combination. Normalised Path Deviation Index (NPDI) was used to identify best-match LCPs generated using different values and their combinations. Original path deviation index can be defined as an average distance between two paths that are compared (Jan et al., 2000); the index however depends on the length of the compared paths, and so, its values are not commensurable among many paths. NPDI overcomes this shortcoming by normalising the index by shortest Euclidian distance between start and end point of a path, as implemented in the *leastcostpath* package by Lewis (Lewis, 2020).

From the start, it became clear that critical slope and topographic variables are insufficient in modelling the roads in the areas with low topographic variability (Fig. 3A). Therefore, it was decided to include additional slope categories. The conductivity values assigned to the slope categories are derived from Tobler's hiking function and reflect the ratio of speed of movement on a given slope to the speed of movement on a slope of 0°. Two models were defined, model 1 more sensitive to the slope and model 2 less sensitive to the slope. Table 1 illustrates the assignment process, Fig. 3B–C shows the conductance surface of Model 1 and its combination with topographic variables. After appropriate conductivity values for critical slope and topographic variables were selected, they were combined with the two models of slope categories to test which is more suitable in the same iterative workflow as previously using NPDI.

Historical lakes were made impassable and marshlands were assigned low conductivity values. The lowest conductivity values were retained in overlapping

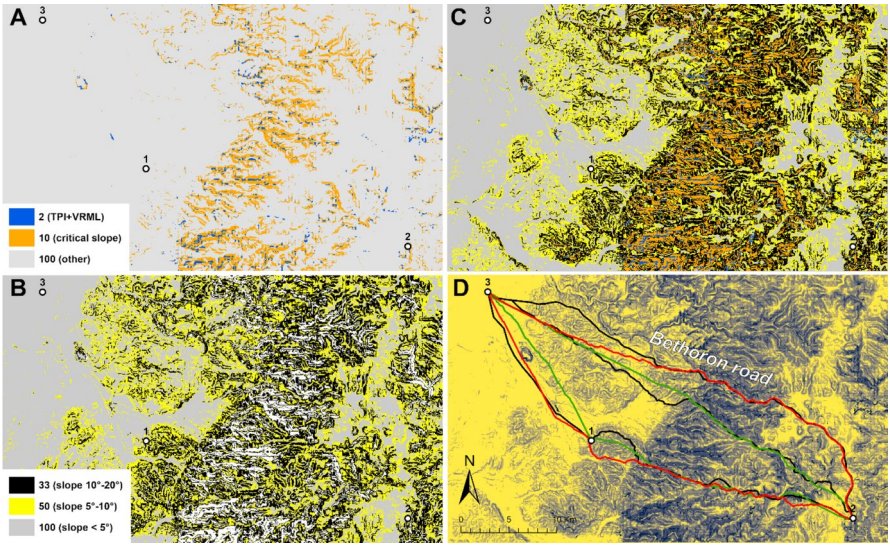

Fig. 3 **A** Conductance surface showing only topographic variables and critical slope in the Jerusalem area test case. **B** Conductance surface showing slope categories of model 1. **C** Conductance surface showing combined conductance surface of model A (more sensitive to topography and slope). **D** Test of assigning the conductivity values with Roman roads (black lines), model A LCPs in red, model B LCPs in green. Sites shown: 1—Emmaus, 2—Jerusalem, 3—Lydda

Table 1 Key to assigning values to slope categories in a conductivity surface based on speeds on different slope in Tobler's hiking function. If speed on 0° slope = 100, then speed on 5° slope is only 73% of the speed on the 0° slope

Slope	Terrain	Ratio of speed of movement based on Tobler's hiking function (0° = 100)	Conductivity values (model 1)	Conductivity values (model 2)
0°–5°	Flat	100–73	100	100
5°–10°	Gentle slopes	73–53	50	
10°–20°	Medium slopes	53–28	33	50

areas. The final conductance surface is isotropic, i.e., direction-independent, assuming equal passability in all directions including the reclassified slope values. The idea was to provide a simple model based on few variables that could be easily evaluated.

Natural Corridors of Movement

Modelling of the natural corridors of movement is adapted from the method proposed by White and Barber (White & Barber, 2012), which is called 'from everywhere to everywhere' (FETE). In each simulation run, a set of 100 random points is generated in the study region, and LCPs are calculated from each point to all other points, resulting in 9,900 LCPs generated in each run. The simulation ends after 50 runs, resulting in a total of 495,000 LCPs. The LCPs are then combined, a frequency value (such as 1 LCP = 1) is calculated, and summarised within a 1000-m neighbourhood and rasterised. The top 1, 5, and 10% of frequency values (i.e., 99, 95, and 90 percentile) in the resulting raster are polygonised, and overlap is calculated for the full dataset of Roman roads (including those roads that are not 'certain'). A 1000-m neighbourhood was selected as it was previously demonstrated that at this scale the effect of movement corridors on the location of Roman sites is most pronounced (Lewis, 2024a).

For comparison with slope-based models, four functions were selected: Tobler (Tobler, 1993), Naismith (Naismith, 1892), Herzog (Herzog, 2013a), and Llobera-Sluckin (Llobera & Sluckin, 2007). Due to computational constraints, simplified FETEs were modelled using only 50 source points and 10 iterations resulting in 24,500 LCPs for each algorithm.

Due to the computationally intensive LCP calculations, the conductance surface was resampled from a ca. 34 m to a 75 m resolution using the majority function in a 4 × 4 window. The majority function was used since it does not interpolate new values in the conductance surface, which is composed of discrete data. The DEM used for slope-based functions was resampled to 75 m using the bilinear function. The LCP modelling, using 16 neighbours (knight's move), was implemented in R using the *leastcostpath* package (Lewis, 2020). The code is available at https://github.com/pazadam/VIA-TARIQ.

Results

Analysis of Topographic Variables

The distribution of slope, TPI maximum, TPI minimum, and VRML values is shown in Fig. 4. Slope and VRML values are both positively skewed. Therefore, the top 99–95 percentile of values was taken to represent the maximum limits for the road location and cutoff values were assigned accordingly. Approximately 98.4% of all road length is located below 20° slope but note the long tail with extreme values up to 39.9°. Calculation of VRML was heavily influenced by an edge effect, where all road segments located within 1 raster cell from the sea (represented by null values) show extremely high values. After these segments were removed from consideration (76 segments, 75.3 km) ca. 95% of roads falls below 0.002332. Both TPI minimum and maximum values are close to the normal distribution, with median values around 0. When taking the minimum value of TPI max (−80.71) and maximum value of TPI min (91.17) as a range of values where Roman roads are typically located, we notice that only ca. 0.45% of road lengths fall outside this range. Table 2 summarises the results and cutoff values used for the conductance surface. We may note that total 93.68% of the road length (3,451.3 km, not including segments affected by the VRML edge effect) fits all three criteria (slope < 20°, TPI −80.71- + 91.17, VRML < 0.002332).

Assigning Conductance Surface Values

The conductance surfaces were created by reclassifying the critical slope, TPI, and VRML rasters based on cutoff values. The topographic variables were combined in a single raster, and the resulting topographic and critical slope rasters were reclassified to be used as conductance surfaces (Fig. 5A–B). Table 3 shows NDPI values separately for topographic and critical slope, for their combination, and for combinations with slope categories. The conductance surfaces were ranked in order of best-fit values for each road. Two combinations of values performed best in modelling the roads in the test case (Fig. 3D): one more sensitive to critical slope and topography (values 10 and 2 respectively), and one less sensitive (values 20 and 10). These two combinations were combined with two slope categories models. From the initial results (Table 3), it seemed that the best model is the one less sensitive to slope categories and topography (using only two slope categories and higher conductivity values for topography and critical slope). However, one of the LCPs produced by a model more sensitive to slope categories, critical slope, and topography in fact very closely follows so-called Bethoron road between Jerusalem and Lydda (Fig. 3D) that was not included in the initial test. After including this road, the ranking of models changed. Nevertheless, as the two best performing models on the one hand performed very similarly and on the other hand seem to capture different roads with different sensitivity to slope categories and topography, it was decided to use both models in FETE modelling. The conductivity values used in both models are

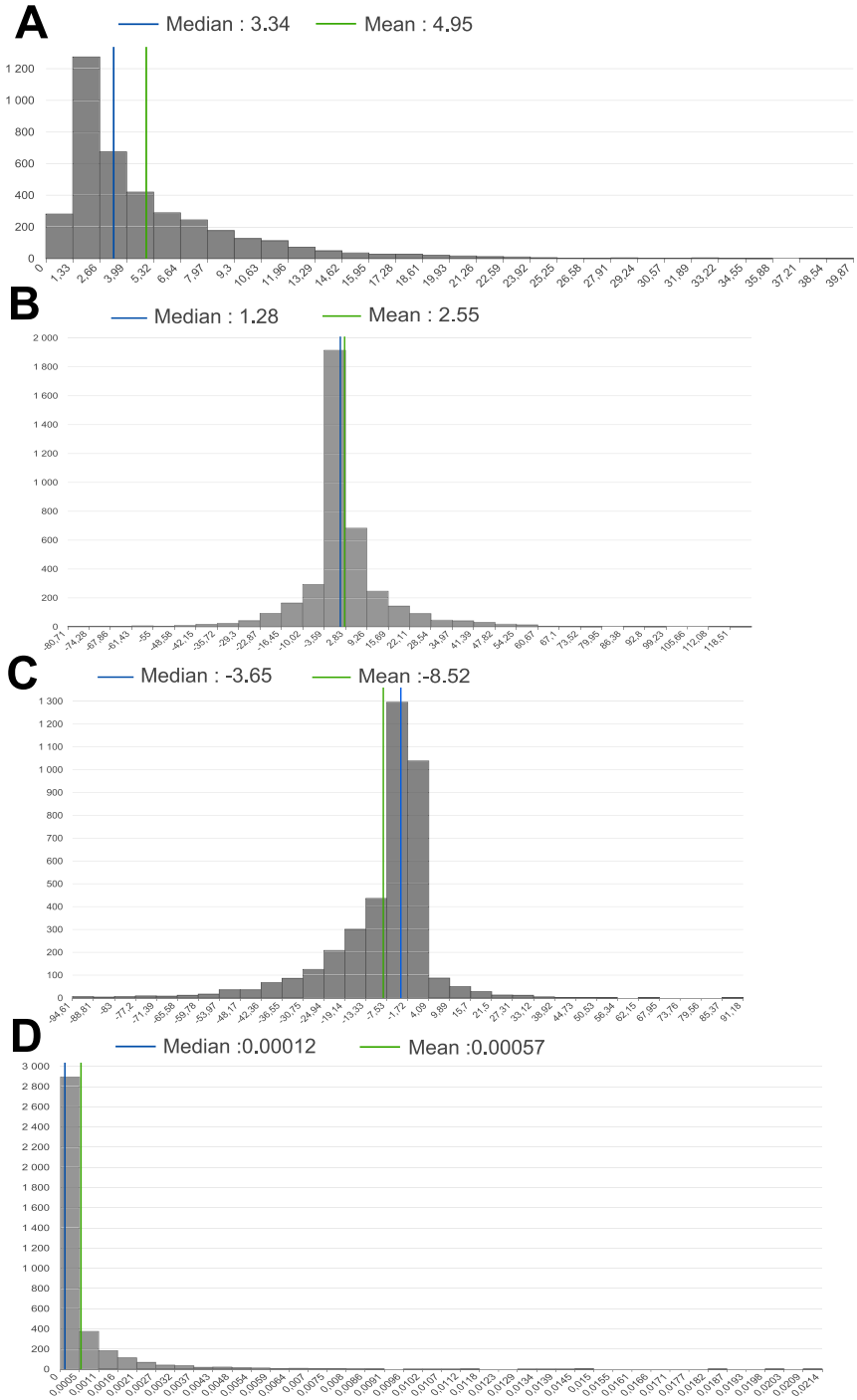

Fig. 4 Distribution of **A** maximum slope, **B** maximum TPI, **C** minimum TPI, and **D** maximum VRML

Table 2 Results of the topographic analysis. Cutoff shows values that were used to assign conductivity values, percentages in parentheses indicate proportion of roads below the cutoff

	Minimum	Maximum	Mean	Median	Cutoff
Slope	0	39.9	4.95	3.34	20 (98.4%)
TPI max	−80.71	124.93	2.55	1.28	−80.71–+91.17 (99.5%)
TPI min	−94.61	91.17	−8.52	−3.64	
VRML max	0	0.021417	0.00057	0.00012	0.002332 (94.4%)

described in Table 4, Model A is more sensitive to slope categories and topography, while Model B is less sensitive to slope categories and topography.

Natural Corridors of Movement

The computed natural corridors of movement, defined as the 99, 95, and 90 percentile frequency of LCPs in the 1000-m neighbourhood, are shown in Fig. 6A–B together with the road network adapted from the *Itiner-e* dataset and overlapping sections of roads. Table 5 summarises the overlap of Roman roads with the corridors for both analysed roads and all roads in the dataset. Most computed corridors run roughly north–south, the orientation of the research region, with few diagonals or east–west corridors. The 99 percentile corridors are divided into several disconnected segments, mostly in the central part of the region. In model A, a long continual section follows the piedmont of the Judaean and Samaritan Highlands from the northern Negev through the coastal plain, passing through Lower Galilee to the Golan Heights. Another follows the central part of the Jordan Valley, meeting the previous corridor in the northern Negev. Another corridor branches in the central Jordan Valley north of the Dead Sea. It ascends the Transjordanian plateau and runs south, bypassing the deep wadis. Finally, a corridor runs around the southern tip of the Dead Sea. The 95 and 90 percentile corridors add connections in the Judaean and Samaritan Highlands, Negev, Transjordan, and upper Jordan Valley. They are more connected, but we see also many parallel routes or wide corridors in plains and valleys. The model B does not significantly differ from model A in where the major corridors are located. Small changes can be observed in flat areas with wider corridors and in the hilly areas, where corridors are not channelled to one route, but some parallel corridors exist. The overlap with Roman roads is not very high at around ~ 24.5% for the 90 percentile corridors, although the 'certain' road segments score slightly better at ~ 27%.

Natural corridors of movement modelled using the four selected slope-based algorithms are presented in Fig. 7. A summary of the overlap and comparison with Model A and Model B is presented in Table 6. The results obtained from all four slope-based algorithms are very similar, with a major continuous corridor running along the length of the Jordan Valley. Parallel north–south corridors in the piedmont of the Judaean and Samaritan highlands in the coastal plain and on the Transjordanian plateau, east of the valleys, are also found in common with the isotropic models. However, fewer corridors are found in the hilly

Table 3 Calibration test for assigning conductivity values to topographic variables, critical slope and slope categories (model 1 and model 2). NPDI values are given for comparison between LCPs and roads. Score is a sum of individual rankings of each conductance raster for a given road (e.g., model2_sl10_topo_02 ranks as third, second, and first best for the three compared roads giving score 3 + 2 + 1 = 6, giving it final ranking 1)

	Emmaus-Jerusalem	Emmaus-Lydda	Jerusalem-Lydda	Score	Final ranking
topo_01	3.9523	9.6489	4.3919	9	4
topo_02	3.9035	9.4824	4.4914	7	1-3
topo_05	3.9042	9.0910	4.5411	7	1-3
topo_10	4.3208	9.0553	4.2579	7	1-3
topo_20	4.6603	9.3856	4.7312	15	5-6
topo_50	4.2567	10.306	4.3562	15	5-6
sl_05	2.8341	8.9652	0.9796	7	1
sl_10	3.2011	9.6821	0.9920	12	5-6
sl_20	2.9764	8.0920	1.0169	8	2
sl_30	3.3201	8.4440	1.0275	12	5-6
sl_50	3.3876	9.4239	0.9321	11	3-4
sl_90	2.5181	10.3188	1.1888	11	3-4
sl_05_topo_02	2.9189	9.1279	1.0178	12	4
sl_05_topo_05	2.5689	10.4803	0.9741	14	5
sl_05_topo_10	3.2343	9.3778	0.9982	17	6
sl_10_topo_02	2.5082	9.3607	0.9711	7	1
sl_10_topo_05	3.1093	10.4615	1.0319	20	7-8
sl_20_topo_02	3.1148	9.4127	1.0478	20	7-8
sl_20_topo_05	2.7700	9.7113	0.8779	10	3
sl_20_topo_10	3.0354	9.0914	0.9326	8	2
model1_sl_10_topo_02	1.1110	8.7715	0.8716	7	2
model2_sl_10_topo_02	2.1126	1.8527	5.4283	9	4
model1_sl_20_topo_10	1.2656	8.0794	0.8173	4	1
model2_sl_20_topo_10	2.1730	1.7988	5.4172	8	3
	Emmaus-Jerusalem	**Emmaus-Lydda**	**Jerusalem-Lydda (Bethoron road)**		
model1_sl_10_topo_02	1.1110	8.7715	6.3889	9	4
model2_sl_10_topo_02	2.1126	1.8527	1.4931	6	1
model1_sl_20_topo_10	1.2656	8.0794	6.3145	8	3
model2_sl_20_topo_10	2.1729	1.7988	1.5100	7	2

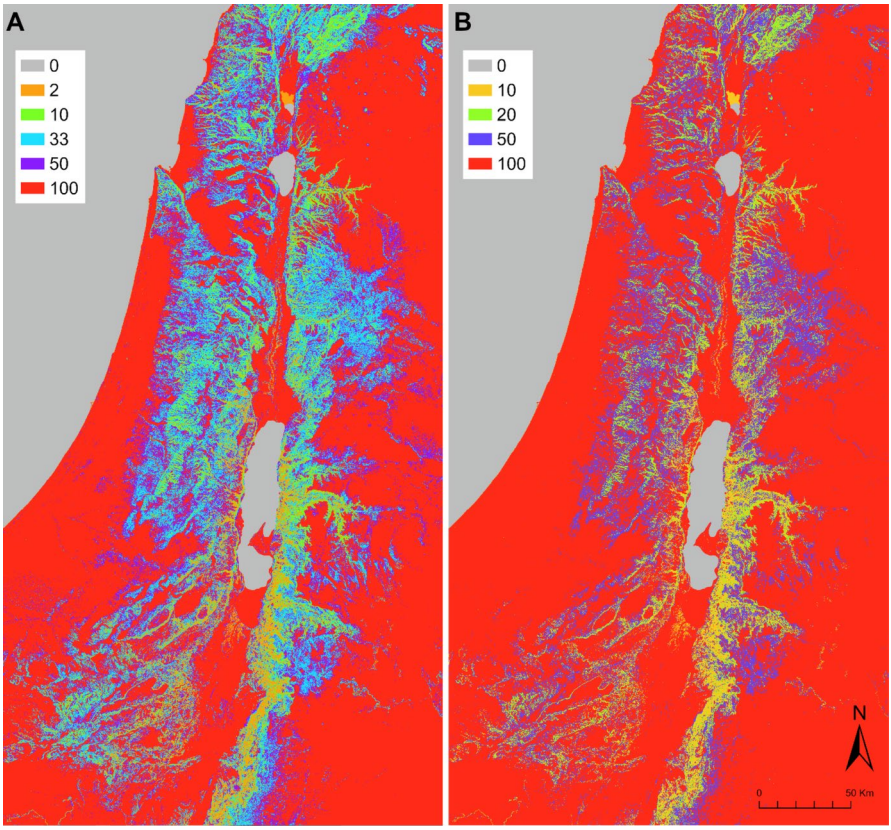

Fig. 5 Conductance surfaces used in the modelling of natural corridors of movement. **A** Model A. **B** Model B

Table 4 Conductivity values assigned in the two models. Model A is more sensitive to slope categories and topography. Model B is less sensitive

Surface	Model A	Model B
0°–5°	100	100
5°–10°	50	
10°–20°	33	50
20° +	10	20
TPI + VRML	2	10
Marshlands (Amuq, al-Ghab, Jabboul, ar-Ruj, Biqqa, Hule)	2	2
Lakes (Gavur Gölü, Amuq, ar-Ruj, Homs Lake, Hule, Dead Sea)	0	0

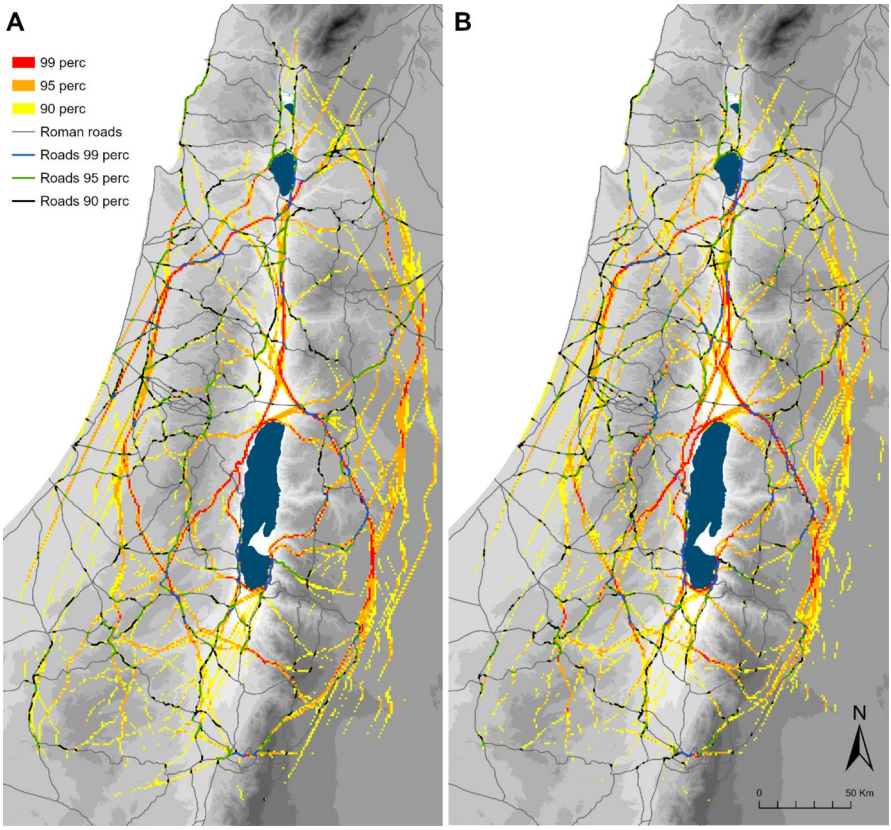

Fig. 6 Modelled natural corridors of movement using isotropic models with overlapping Roman roads. **A** Model A. **B** Model B

Table 5 Overlap between 99, 95, and 90 percentile frequency values of the modelled natural corridors of movement and the road dataset (analysed roads are also contained in 'All roads')

Model A	**Analysed roads**	**All roads**
99 percentile	4.39%	3.47%
95 percentile	15.04%	13.65%
90 percentile	26.55%	24.52%
Model B	**Analysed roads**	**All roads**
99 percentile	5.19%	3.98%
95 percentile	15.59%	14%
90 percentile	27.09%	24.29%

areas across the case study region. The overlap with Roman roads is weaker for all four algorithms, with maximum overlap in 90 percentile frequency category between ~ 14.7 and 17%, so 7–10% less than in the isotropic models.

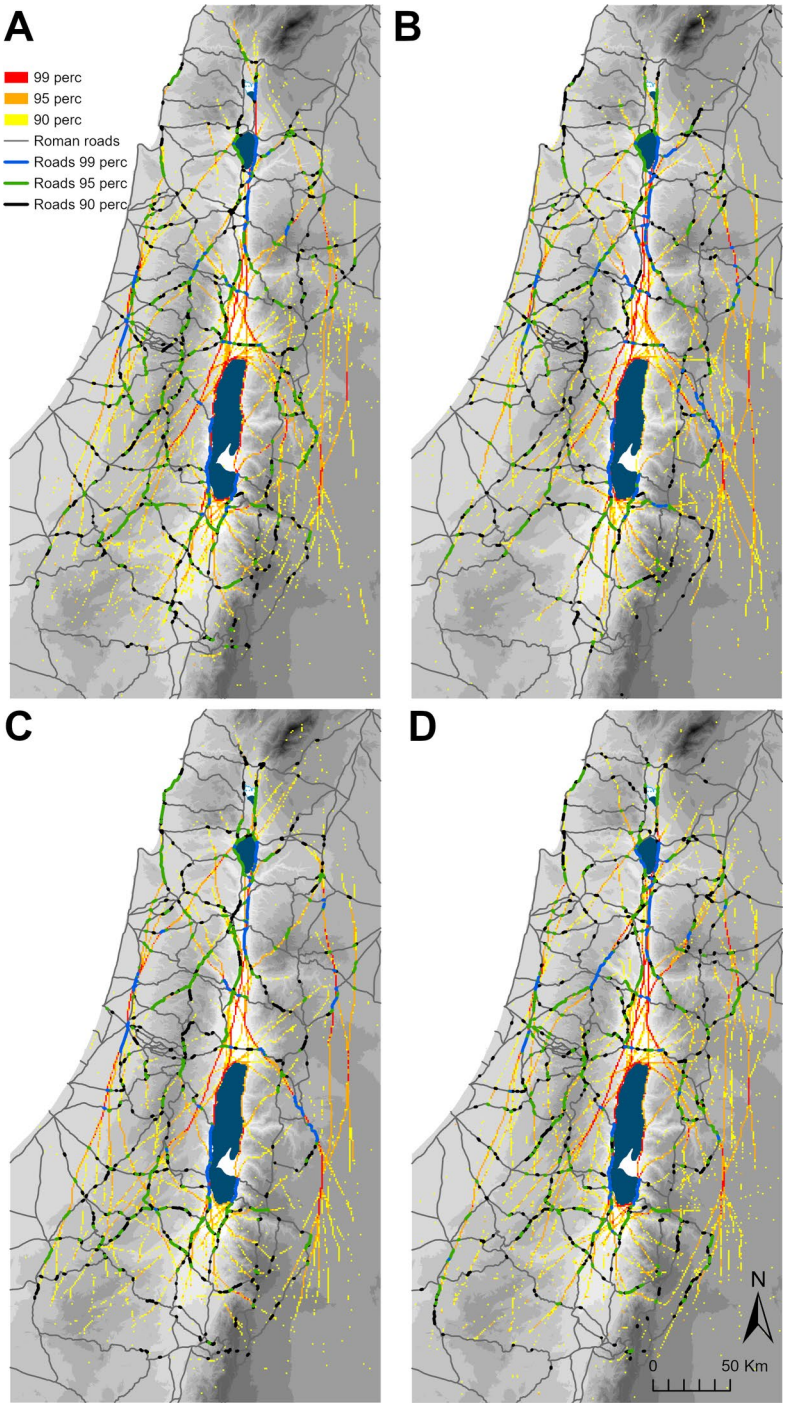

Fig. 7 Modelled natural corridors of movement using slope-based functions with overlapping Roman roads. **A** Tobler. **B** Naismith. **C** Herzog. **D** Llobera-Sluckin

Table 6 Overlap between 99, 95, and 90 percentile frequency values of the modelled natural corridors of movement and the road dataset using both slope-based algorithms and isotropic models A and B

Model	99 percentile	95 percentile	90 percentile
Model A	3.47%	13.65%	24.52%
Model B	3.98%	14%	24.29%
Tobler	2.27%	10.01%	17.03%
Naismith	2.65%	8.52%	14.74%
Herzog	2.52%	9.30%	16.54%
Llobera-Sluckin	2.19%	9.65%	17.04%

Discussion

Influence of Elevation Data

The resolution and quality of the DEM can influence both the calculation of the topographic variables and the length and shape of the calculated LCPs (Herzog & Posluschny, 2011). The actual length and shape of the LCPs is of little interest in modelling the movement corridors as LCPs are aggregated in a specified neighbourhood (1000 m in this case), and therefore it may be disregarded.

We may instead explore the sensitivity of the slope limit calculation in the topographic analysis by comparing the results obtained from the FABDEM to other DEMs. The following DEMs were selected for the comparison: SRTM 1-arc second (Earth Resources Observation And Science (EROS) Center 2017), ASTER GDEM v3 (NASA/METI/AIST/Japan Spacesystems and U.S./Japan ASTER Science Team 2019), EDTM (Ho et al., 2025), and Copernicus GLO-90 (European Space Agency and Airbus 2022) which should show more smoothing of the values due to its larger resolution. Figure 8 shows histogram of maximum slope values calculated for the certain road segments using these DEMs with 99 percentile marked. We may see that Copernicus is indeed smoothed and tends to have lower values overall, while SRTM and especially ASTER are skewed to higher values. FABDEM and EDTM are comparable both with 99 percentile value around 22°. Both FABDEM and EDTM are supposed to have improved vertical accuracy and mean root square error compared to other global DEMs which often include unfiltered buildings and vegetation cover. The selection of 20° slope as un upper limit for road location (98.5 percentile for FABDEM, 98 percentile for EDTM) seems to accord well with data from these two improved global DEMs.

Natural Corridors of Movement

The calculated natural corridors of movement clearly illustrate several issues with the proposed simple isotropic model. In the first place, the selected topographic variables and their limits clearly did not provide enough friction over large regions with little topographic variability, e.g., the coastal plain or the Transjordanian plateau. In these places, LCPs tend to follow straight lines over long distances, and many run

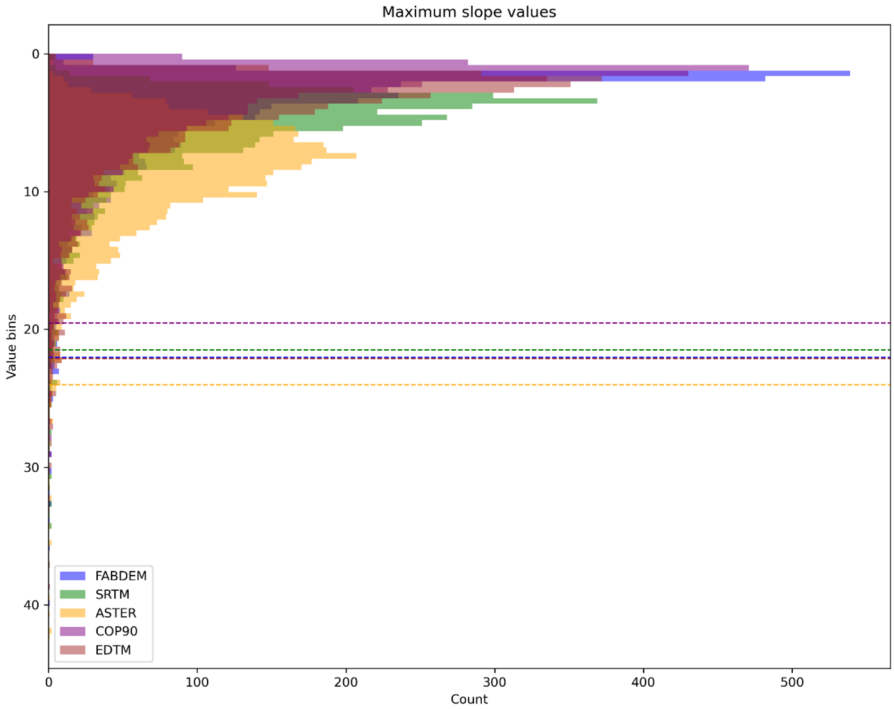

Fig. 8 Histogram of maximum slope values calculated from different DEMs

parallel to each other as the conductance surface does not force them to converge in a single corridor. This effect can be observed also in areas of gentle topography such as piedmonts and low hills where LCPs converge either in discrete places (crossroads) or along short segments which causes breaking of the corridors into non-continuous segments. Such regions with gentle topography would necessitate creation of a different model based on additional variables. Part of this region lies in (semi-) arid zone so availability of water sources would be more influential variable than topography (Manière et al., 2021). Another issue is that the shape of the computational region, which roughly extends N-S, makes LCPs and corridors to have the same orientation with few E-W and diagonal connections, typically in places where the LCPs are forced through mountain passes or ridgeways between the coast and the hinterland. Edge effect influenced mainly north-eastern and southern extremes of the region resulting in poor overlap with Roman roads. Although in the case of the north-eastern corner the contributing factor of the poor overlap is the flat region with little topographic variability.

Despite these limitations, it is possible to observe some interesting coincidences between modelled corridors, Roman roads, and location of important nodes (Roman sites). The model generally well captures potential movement corridors in hilly and mountainous zones, and so e.g., many Transjordanian ascents (Aqabat Fiq, Gadara, Mahatta, Kathrabba pass, Petra – Naqb el-Aqab) and Negev ascents (Maaleh Tamar, Maaleh Aqravim) are fully or partially captured (Fig. 9A–C) and couple of highland

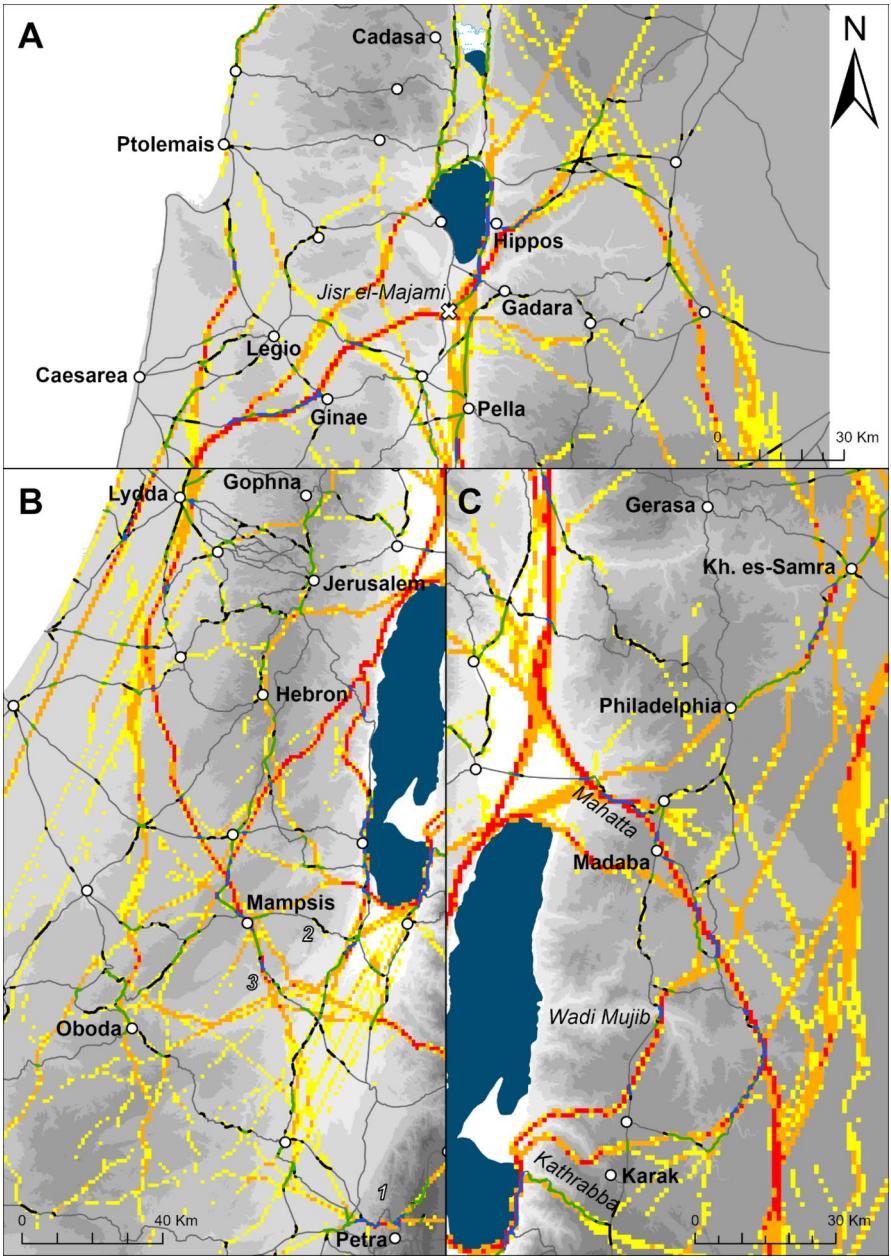

Fig. 9 Details of the overlap between corridors, Roman roads, and sites mentioned in the text. **A** Northern part of the region. **B** Judaea and Negev. **C** Central Transjordan. 1—Naqb el-Aqab, 2—Maaleh Tamar, 3—Maaleh Aqravim

roads (e.g., Bethoron road, 'watershed road' between Gophna and Hebron, etc. Figure 9B). The model coincides well mainly with Roman roads going N-S along the coastal plain and along the length of the Rift Valley (Fig. 9B–C). *Via Nova Traiana* is well captured between Khirbet es-Samra to Philadelphi and then south to Wadi Mujib where crossing of this wadi is captured very accurately (Fig. 9C). Among the places located on the major corridors are several key Roman sites, such as Hippos on the crossroad of corridors through the Jordan Valley and the Golan Heights, Pella, Philadelphia-Amman, Madaba, Characmoba-Kerak, Petra in Transjordan and Antipatris, Lydda, Jerusalem in Judaea, and Mampsis and Oboda on the Incense route in the Negev, with many more minor settlements and forts. A major corridor crosses bridge at Jisr el-Majami between Scythopolis and Gadara (Fig. 9A). There are also disconnections between some Roman sites and modelled corridors. For example, there is no major corridor in Meggido-Legio area, which is historically attested as major crossroad of international routes (Isaac & Roll, 1982). Instead, major corridor runs through Ginae-Jenin and Narbata. Likewise, Gerasa is disconnected from all corridors in Transjordan despite being a crossroad of several main roads in the Roman period (Mittmann, 1970). Caesarea was chosen as the main provincial harbour and capital rather for its maritime connectivity than for terrestrial mobility.

The movement corridors modelled using slope-based algorithms tend to follow valleys and flat areas more systematically, which results in fewer corridors in the hilly and mountainous areas. This 'channelling' of LCPs into few corridors is the main reason for low overlap with Roman roads. Moreover, part of the overlap is caused by perpendicular crossings, which are accidental and a poor match.

Explaining the Location of Roman Roads

Nevertheless, the overlap between modelled corridors and the Roman roads in the isotropic models A and B is only ~ 25%. It also needs to be admitted here that part of this overlap is caused by the accidental perpendicular overlap, although it seems to be less prevalent than with the slope-based functions. On the other hand, some modelled corridors follow the general direction of a road and run parallel to it but do not overlap within the given neighbourhood. This might be partly caused by the size of the neighbourhood used to aggregate the calculated LCPs to corridors (1000 m), although not all corridors are limited to 1000 m width, as multiple LCPs can converge close to each other in an area of several kms creating a wide corridor. The low overlap is likely caused by a combination of two factors: (1) models with selected topographic variables do not perform well in flat regions, as was discussed above; and (2) the scale of the analysis does not permit capturing enough potential natural corridors of movement. Additionally, the spatial accuracy of certain roads in the dataset might be too low in certain places, although given the width of the corridors, this should not be an issue. Or perhaps we may reverse our perspective and state that the identified topographic limits of the Roman road locations offer a wide niche in the landscape where Roman roads can be located, and the road layout (in the sense of an actual realised route in the landscape, Watteaux, 2017) is a result of additional processes related to the development of settlement structures, economy, persistency

(building new roads is more expensive than maintaining existing roads), and local circumstances (Earle, 2011; Llobera, 2020; Pažout et al., 2024).

Conclusion

In this article I aimed to evaluate the importance of the selected topographic variables on the location (layout) of the Roman roads in the Roman Levant. A key aspect of the proposed methodology is that it is building the model parameters inductively from the analysis of a large dataset of Roman roads. This approach allows us to define topographic constraints of movement more systematically and rigorously, assuming that Roman roads are valid proxies reflecting past human movement and its limitations. This analysis might be expanded to other provinces of the Roman Empire where a similar set of highly spatially accurate road segments is present in the *Itiner-e* dataset. This would allow us to test the main conclusions of this article and to propose a general model for the Roman Empire, or perhaps a set of general models for different geographical zones of the Roman Empire, incorporating hydrological regimes, climatic data and additional topographic variables. This methodological approach, however, might be applied to all pre-modern road systems, where appropriate detailed road data exist. I would strongly advocate for such an approach since it starts with the archaeological data from which it inductively builds basic premises that help us post-dict (explain) the material remains and patterns of past movement.

The model of the natural corridors of movement built on the analysed topographic limits of the Roman roads showed that the Roman roads do not overlap with the corridors (i.e., with places in the landscape that are more conductive to channel movement). While there are several explanations for this related to the algorithm (low explanatory value in places of low topographic variability), quality of the Roman road data (low spatial accuracy of parts of the dataset and potential missing data), the inevitable conclusion is that the selected topographic variables (critical slope, range of TPI values, critical value of VRML) do not explain the location of the Roman roads well and their influence on the layout of Roman roads, while it existed, is limited. The fact that these variables had an influence on layouts of Roman roads is corroborated by comparison of isotropic models with the slope-based functions which showed that coincidence of Roman roads with isotropic models is higher, and combination of slope categories and topographic variables better explain locations of Roman roads than slope only. If slope algorithms explain ca. 14.5–17% of the locations and Models A and B ca. 25%, we may surmise that topographic variables explain additional 7–10% of locations after slope. The layout of the Roman roads and of the Roman road network therefore must be explained by other factors not considered in this research. Besides topographic variables, possible water availability might provide better explanation in the (semi-)arid regions of the Roman Near East, but it will probably not be decisive variable in the Mediterranean climate zone with better availability of water. The main drivers behind the decision process leading to the creation of road layout therefore must be anthropic, cultural variables, such as settlement patterns, long- and short-distance exchange networks, persistency

of the road network, state interventions (for strategic or military reasons), land use and land ownership, etc. Arguably, many of these variables are also related to topographic and environmental variables, but the way how people manipulated their environment might be highly specific. Alternatively, the assumption that the Roman roads were globally optimised to minimise cost could be wrong, but as the results show there is some degree of optimization and the variables not considered in this research might potentially explain the rest. In light of this conclusion, it seems that a model combining topographic and anthropic variables would be better suited to explain the layout of Roman roads, and for this least-cost path networks might offer suitable methodology (Carroll & Carroll, 2023; Groenhuijzen & Verhagen, 2017; Herzog, 2023; Prignano et al., 2019).

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Data Availability The R code used for modelling LCPs and NPDI comparison is stored at github repository (https://github.com/pazadam/VIA-TARIQ). The raster data (DEM, TPI and VRML), dataset of analysed roads with attached values of maximum slope, TPI and VRML, and resulting FETE LCP frequency rasters are stored at Zenodo repository (https://doi.org/10.5281/zenodo.17953712).

Declarations

Competing Interests The authors declare no competing interests.

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